YAORP (Yet Another Open Reflow Plate)
– Controller Board –

Power – 240AC to 5VDC

File: sch_power_ACto5VDC.kicad_sch

Power – 3v3DC

File: sch_power_3v3DC.kicad_sch

IO – Thermocouple

File: sch_io_thermocouple.kicad_sch

MCU – ESP32 S3 Wroom 1

File: sch_mcu_esp32s3wroom1.kicad_sch

IO – PTC Plate

File: sch_io_ptc.kicad_sch

IO – Cooling Fan

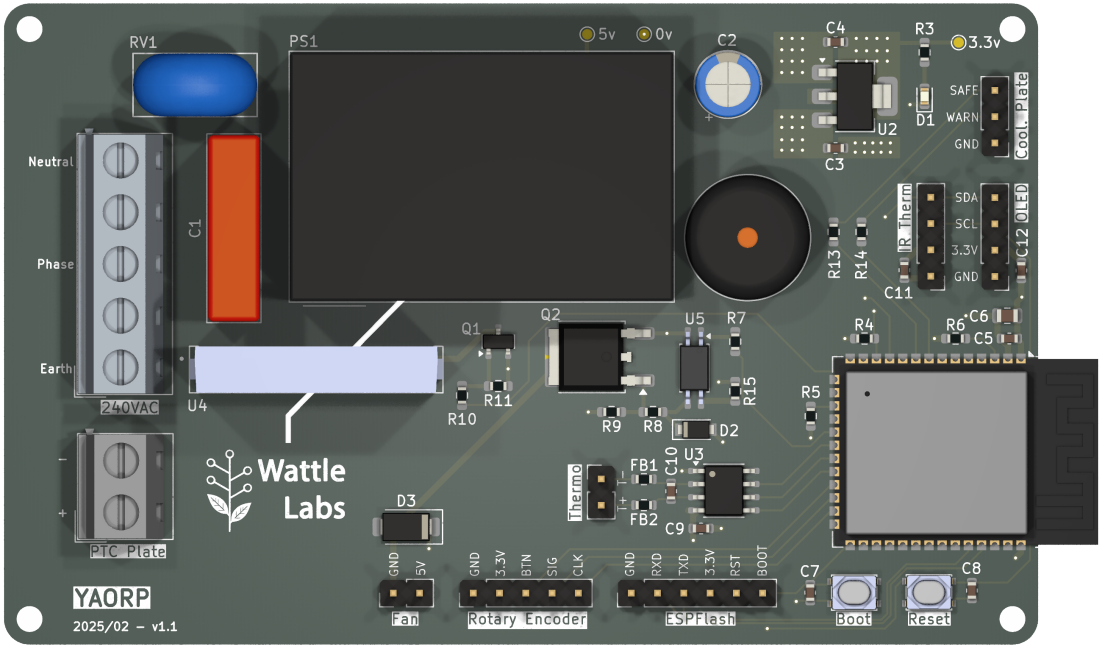
File: sch_io_fan.kicad_sch

IO – Misc

File: sch_io_misc.kicad_sch

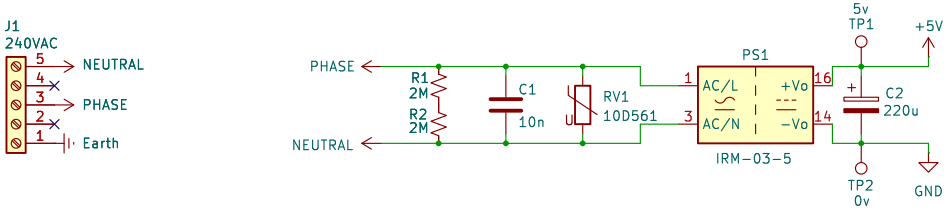
PCB – Mounts

File: sch_pcb_mounts.kicad_sch



Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /			
File: Board1.kicad_sch			
Rev: 1.1	Date: 2025-02-12	Id: 1/9	KiCad E.D.A. 8.0.8

Mains to 5v DC Power Supply

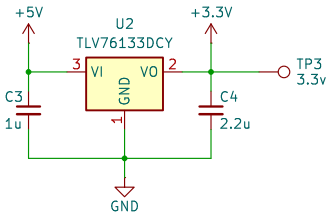


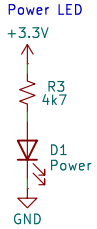
Design Notes:

Mains side 2A fuse is intended to be offboard, being built into IEC power receptable

Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /Power – 240AC to 5VDC/ File: sch_power_ACto5VDC.kicad_sch			
Rev: 1.1	Date: 2025−02−12	Id: 2/9	KiCad E.D.A. 8.0.8

3.3v DC Linear Regulator (LDO)





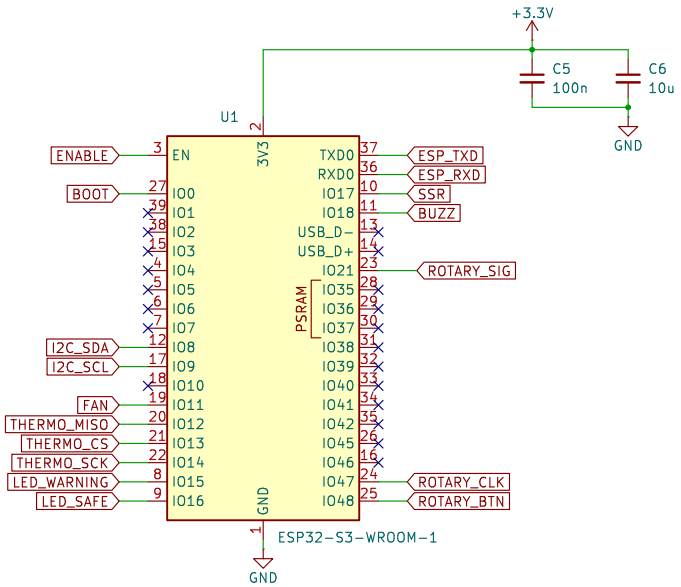
V_{IN}: 2.5V–16V (20V abs max)

V_{OUT}: fixed 3.3V, max 1A

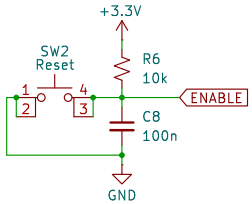
I_Q: 60uA (~1.5uA in shutdown)

Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /Power – 3v3DC/ File: sch_power_3v3DC.kicad_sch			
Rev: 1.1	Date: 2025-02-12	Id: 3/9	KiCad E.D.A. 8.0.8

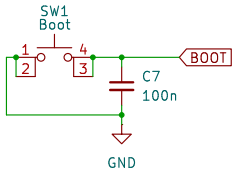
ESP32-S3-WROOM-1 Module



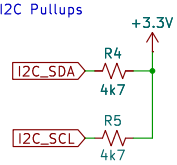
ESP32 Reset



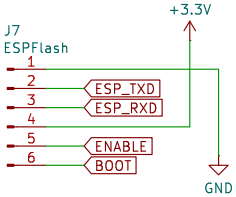
ESP32 Boot



I2C Pullups



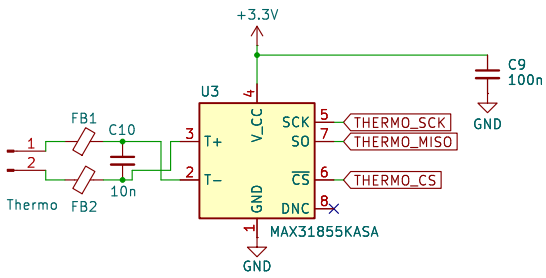
ESPFlash Connector



This could be removed in favour of a USB connector however cost uplift and slightly more complex to route. The firmware is unlikely to be updated often AND could be done OTA.

Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /MCU - ESP32 S3 Wroom 1/ File: sch_mcu_esp32s3wroom1.kicad_sch			
Rev: 1.1	Date: 2025-02-12	Id: 4/9	KiCad E.D.A. 8.0.8

Type K Thermocouple (MAX31855) Driver

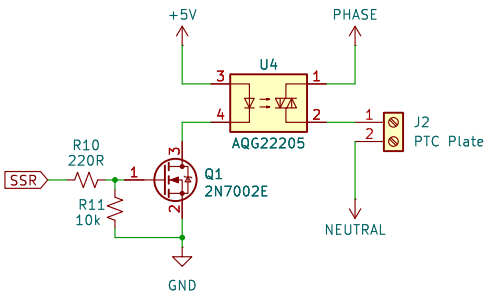


Design Notes:

1. Capacitors are those recommended in MAX31855 datasheet.
2. Firmware PoC tested with MAX6675 but MAX31855 is newer part (NOT pin compatible)
3. Ferrites added to deal with potential EMI issues (MAX31855 is sensitive).

Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /IO – Thermocouple/ File: sch_io_thermocouple.kicad_sch			
Rev: 1.1	Date: 2025-02-12	Id: 5/9	KiCad E.D.A. 8.0.8

Solid State Relay to drive PTC Plate

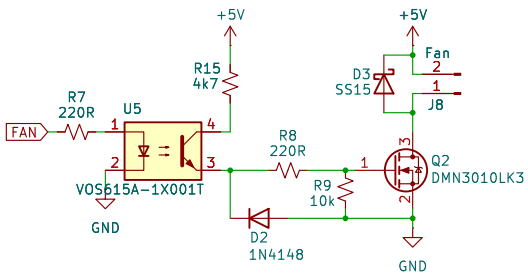


Design Notes:

1. SSR is a 5V drive thus MOSFET required to enhance ESP32 GPIO.

Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /IO – PTC Plate/ File: sch_io_ptc.kicad_sch			
Rev: 1.1	Date: 2025-02-12	Id: 6/9	KiCad E.D.A. 8.0.8

Cooling Fan Driver



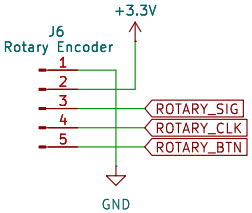
Design Notes:

1. Optocoupler used to buffer 3.3v to 5v drive for MOSFET

Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /IO – Cooling Fan/ File: sch_io_fan.kicad_sch			
Rev: 1.1	Date: 2025-02-12	Id: 7/9	KiCad E.D.A. 8.0.8

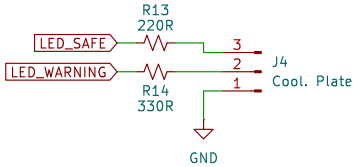
Misc I/O

Rotary Encoder Connector



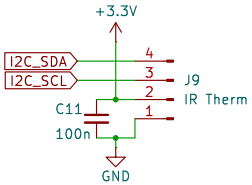
Design Note: Encoder module includes pullups

Cooling Plate LED Connector

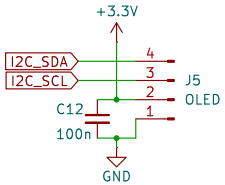


Note: LEDs live on a seperate schematic/PCB (board 2)

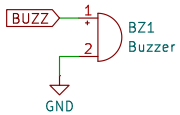
Cooling Plate IR Thermometer Connector



OLED Connector



Buzzer



Title: YAOSRP Reflow Plate Controller			Wattle Labs
Sheet: /IO – Misc/ File: sch_io_misc.kicad_sch			
Rev: 1.1	Date: 2025-02-12	Id: 8/9	KiCad E.D.A. 8.0.8

Mounting Holes

H1
MountingHole

H2
MountingHole

H3
MountingHole

H4
MountingHole

Title: YAOSRP Reflow Plate Controller			
Sheet: /PCB – Mounts/ File: sch_pcb_mounts.kicad_sch			Wattle Labs
Rev: 1.1	Date: 2025-02-12	Id: 9/9	KiCad E.D.A. 8.0.8